

Hybrid Model Predictive and Nonlinear Feedback Control for Chaos Suppression and MPPT Enhancement in DFIG-Based Wind Energy Systems

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Abstract—Doubly fed induction generator (DFIG)-based wind turbines are strongly nonlinear electromechanical systems that can exhibit chaotic dynamics under perturbed parameters and adverse operating conditions, leading to degraded power quality and compromised stability. This paper analyses a third-order rotor-current/speed model of the DFIG with a physically consistent parameter set, characterizes its equilibria, dissipativity and Lyapunov exponents, and proposes a hybrid controller acting only on the two rotor voltages. An outer speed loop generates the torque-current reference, nonlinear feedback compensation cancels the coupled current dynamics, and a constrained receding-horizon Model Predictive Control (MPC) with integral action, whose terminal weight and gain are obtained from a linear matrix inequality, handles the converter voltage limits. Simulations show that the controller suppresses the chaotic oscillations without steady-state error, and tracks Maximum Power Point Tracking (MPPT) references under turbulent wind with a speed error about 146 times and a power error about 7.7 times lower than a classical cascaded PI controller.

Index Terms—Doubly-Fed Induction Generator (DFIG), Wind Energy Systems, Chaotic Dynamics, Nonlinear Dynamics, Maximum Power Point Tracking (MPPT), Hybrid Control, Model Predictive Control (MPC), Nonlinear Feedback Control, Robust Control.

I. INTRODUCTION

The escalating global population coupled with increasing electricity demand has accelerated the depletion of non-renewable energy resources at unprecedented rates [1]. This critical challenge necessitates the exploration and deployment of alternative, sustainable, and environmentally-friendly energy sources, particularly wind energy, to mitigate environmental degradation and address energy security concerns [2], [3].

Wind energy has emerged as a pivotal renewable energy technology in recent decades, experiencing substantial growth in both global investments and installed capacity [4]. Recent statistics indicate that global wind power capacity expanded by a record 117 GW in 2024, elevating the cumulative worldwide installed capacity to approximately 1,136 GW [5]. This remarkable expansion has been driven by supportive policy frameworks, technological innovations, and declining leveled cost of energy, positioning wind power as one of the fastest-growing renewable energy sources worldwide [6].

Among various wind turbine generator configurations, the doubly fed induction generator (DFIG) has become the dominant technology in large-scale onshore wind farms, primarily due to its partial-scale back-to-back power converter architecture that enables flexible control of active and reactive power with reduced converter ratings [7], [8]. This configuration allows the rotor-side converter (RSC) and grid-side converter (GSC) to independently regulate rotor currents and grid-injected currents, respectively, thereby achieving decoupled control of electromagnetic torque, stator active power, and reactive power over a wide operational speed range [9].

The economic advantage of requiring only 25–30% rated power converters compared to full-scale converter systems, combined with superior power quality and grid-support capabilities, has made DFIG the preferred choice for variable-speed wind energy conversion systems (WECS) [10].

Despite extensive research on DFIG-based WECS including dynamic modeling, vector control strategies, grid-fault ride-through capability, and ancillary services provision, recent studies have revealed that these systems can exhibit complex nonlinear dynamic behaviors under certain operating conditions [11]–[13]. The inherent nonlinearity of the induction machine, coupled with the interaction between mechanical and electrical subsystems and the coupling between stator and rotor flux linkages, can lead to oscillatory transients and, under severe conditions, bifurcations and quasi-chaotic responses [14], [15].

These phenomena become particularly pronounced under severe grid disturbances, highly turbulent wind speed profiles, parameter uncertainties (such as machine detuning), and converter saturation effects, all of which may degrade power quality, stress mechanical components, and compromise system stability [16], [17].

The emergence of chaotic behavior in DFIG systems manifests as irregular power oscillations, torque ripple, and complex trajectories in the state space, which are detrimental to both electrical and mechanical performance [18]. Such dynamics are especially critical in the context of stringent grid-code requirements for low-voltage ride-through (LVRT), frequency support, and reactive power support [19], [20].

Consequently, the analysis, mitigation, and control of nonlinear and potentially chaotic dynamics in DFIG-based wind turbines have become important research priorities for ensuring stable operation under wide-ranging operating scenarios and maximizing the contribution of wind energy to global electricity supply.

Motivated by these challenges, this paper investigates the control of a variable-speed wind turbine equipped with a DFIG that exhibits chaotic-like behavior induced by machine parameter deviations and adverse operating scenarios, which can severely affect system stability and grid-injected power quality. A comprehensive nonlinear mathematical model is first developed to capture the intricate coupling among rotor currents, electromagnetic torque, and mechanical speed, and to identify the conditions under which complex or chaotic regimes emerge when the system is subjected to significant disturbances, such as abrupt wind-speed variations or degraded parameter settings. Moreover, the system is subjected to a highly realistic wind-speed profile to evaluate its ability to maintain stability and drive its dynamics toward accurate wind-path tracking, as assessed through the tracking error. Building on this framework, an innovative hybrid control architecture is then proposed, combining Model Predictive Control (MPC) with nonlinear state-feedback compensation to suppress chaotic dynamics, enhance robustness to uncertainties, and achieve high-precision tracking of reference rotor current and speed trajectories derived from Maximum Power Point Tracking (MPPT) objectives.

II. WIND TURBINE MODELING

The aerodynamic power captured by a horizontal-axis wind turbine can be written as [21]–[23]

$$P_a = \frac{1}{2} \pi \rho R^2 C_p(\lambda) v^3 \quad (1)$$

where ρ is the air density, R is the rotor radius, v is the wind speed, and $C_p(\lambda)$ is the power coefficient, which depends on the tip-speed ratio λ .

The tip-speed ratio is defined by

$$\lambda = \frac{R\omega_r}{v} \quad (2)$$

with ω_r the rotor (mechanical) angular speed.

The corresponding aerodynamic torque applied to the shaft is obtained from

$$T_a = \frac{P_a}{\omega_r} = \frac{1}{2} \pi \rho R^2 C_p(\lambda) \frac{v^3}{\omega_r} \quad (3)$$

The rotational dynamics of the wind turbine-generator set are described by

$$J \frac{d\Omega_m}{dt} = T_a - T_e - D\Omega_m \quad (4)$$

where J is the total inertia, Ω_m is the mechanical rotor speed, T_e is the electromagnetic torque produced by the generator, and D denotes the viscous friction coefficient.

Relating mechanical and electrical rotor speeds by

$$\omega_r = n_p \Omega_m \quad (5)$$

with n_p the number of pole pairs, and using $\frac{d\Omega_m}{dt} = \frac{1}{n_p} \frac{d\omega_r}{dt}$, (4) can be rewritten in electrical speed form as

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - T_e - \frac{D}{n_p} \omega_r \quad (6)$$

For a DFIG expressed in the stator dq reference frame, the electromagnetic torque is given by

$$T_e = \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (7)$$

where ψ_{ds}, ψ_{qs} are the stator flux components and i_{ds}, i_{qs} are the corresponding stator current components.

Substituting (7) into (6) yields the final electromechanical dynamic equation of the rotor:

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) - \frac{D}{n_p} \omega_r \quad (8)$$

III. CHAOTIC DOUBLY-FED INDUCTION GENERATOR

According to (8) and the stator-flux/rotor-current coupling of the DFIG, a third-order state equation with i_{dr} , i_{qr} , and ω_r as state variables can be obtained [24]:

$$\begin{aligned} \dot{i}_{dr} &= \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{dr} + (\omega_s - \omega_r) i_{qr} \\ &\quad - \frac{\omega_r L_m \psi_s}{\sigma L_r L_s} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{ds} + \frac{1}{\sigma L_r} u_{dr} \\ \dot{i}_{qr} &= -(\omega_s - \omega_r) i_{dr} + \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{qr} \\ &\quad + \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{qs} + \frac{1}{\sigma L_r} u_{qr} \\ \dot{\omega}_r &= \left(\frac{n_p}{J} \right) \left(\frac{3n_p L_m \psi_s}{2L_s} i_{dr} - \frac{D}{n_p} \omega_r - T_L \right) \end{aligned} \quad (9)$$

We now rewrite (9) by introducing the following state variables:

$$x_1 = i_{dr}, \quad x_2 = i_{qr}, \quad x_3 = \omega_r.$$

Let the parameters $a_1, a_2, \dots, a_8$ be defined accordingly. The resulting nonlinear system can then be expressed in the compact state-space form

$$\begin{cases} \dot{x}_1 = a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + (a_3 u_{ds} + a_4 u_{dr}), \\ \dot{x}_2 = -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + (a_3 u_{qs} + a_4 u_{qr}), \\ \dot{x}_3 = a_6 x_1 - a_7 x_3 - a_8. \end{cases} \quad (10)$$

The coefficients appearing in (10) are given by

$$\begin{aligned} a_1 &= -\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r}, \quad a_2 = -\frac{L_m \psi_s}{\sigma L_r L_s}, \quad a_3 = -\frac{L_m}{\sigma L_s L_r}, \\ a_4 &= \frac{1}{\sigma L_r}, \quad a_5 = \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r}, \quad a_6 = \frac{3n_p^2 L_m \psi_s}{2J L_s}, \\ a_7 &= \frac{D}{J}, \quad a_8 = \frac{n_p}{J} T_L, \end{aligned}$$

with $\sigma = 1 - L_m^2/(L_s L_r)$. The stator flux is aligned with the q axis, so that $u_{ds} = -\omega_s \psi_s$ and $u_{qs} = 0$.

TABLE I
MAIN PARAMETERS OF THE CHAOTIC DFIG SYSTEM

Parameter	Value	Parameter	Value
R_s	$5.0536 \times 10^{-5} \Omega$	n_p	2
R_r	$1.8781 \times 10^{-4} \Omega$	J	$31.18 \text{ kg} \cdot \text{m}^2$
L_s	0.0833 H	D	$71.85 \text{ N} \cdot \text{m} \cdot \text{s/rad}$
L_r	0.0844 H	T_L	$-11337 \text{ N} \cdot \text{m}$
L_m	0.0811 H	u_{dr}	2.539 V
σ	0.0645	u_{qr}	-2.936 V
ω_s	$100\pi \text{ rad/s}$	u_{ds}	-563.4 V
ψ_s	1.7933 Wb	u_{qs}	0 V

The parameters listed in Table I are physically consistent: $L_s, L_r > L_m$ and $\sigma = 0.0645 \in (0, 1)$. They give $a_1 = -0.0433$, $a_2 = -320.83$, $a_3 = -178.91$, $a_4 = 183.76$, $a_5 = 0.1946$, $a_6 = 0.3360$, $a_7 = 2.3044$, $a_8 = -727.20$. The constant terms of (10) are therefore $a_3 u_{ds} + a_4 u_{dr} = 1.0126 \times 10^5$ and $a_5 + a_3 u_{qs} + a_4 u_{qr} = -539.32$.

System (10) has three equilibrium points, all unstable:

- $E_1 = (-64.94, -379.12, 306.11)$, a saddle-focus with eigenvalues $1.462 \pm 7.673j$ and -5.315 ;
- $E_2 = (-9.97, -12442.5, 314.12)$, a saddle with eigenvalues $-65.00, 62.65$ and -0.043 ;
- $E_3 = (55.50, -272.00, 323.67)$, a saddle-focus with eigenvalues $0.547 \pm 10.511j$ and -3.486 .

The divergence of the vector field is constant and negative, $\nabla \cdot f = 2a_1 - a_7 = -2.391$, so the system is dissipative.

With these parameters, the dynamical system exhibits complex nonlinear behavior. The corresponding chaotic attractors generated from the initial condition $x(0) = (0.3, 0.6, 3.6)^T$ are illustrated in Fig. 1.

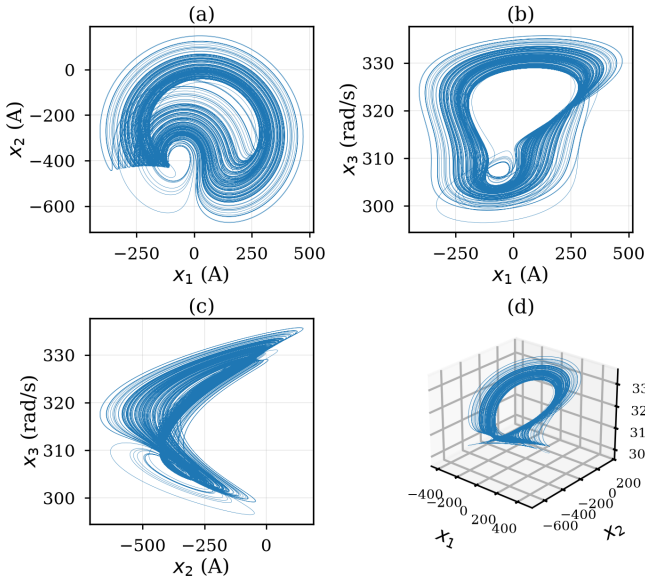


Fig. 1. The chaotic attractors: a) The projection of the chaotic attractor in plan $x_1 - x_2$; b) the projection of the chaotic attractor in plan $x_1 - x_3$; c) the projection of the chaotic attractor in plan $x_2 - x_3$; d) 3D trajectory of the system.

The phase portraits presented in Fig. 1 illustrate the complex nonlinear behavior of the DFIG system. The trajectory winds around the two saddle-foci E_1 and E_3 . The rotor speed remains positive and close to synchronism ($296 \leq \omega_r \leq 336 \text{ rad/s}$), while the rotor currents oscillate irregularly between about -670 and $+470 \text{ A}$. The Lyapunov exponents, computed with the QR-based variational method, are $(0.202, -0.010, -2.582)$. Their sum equals the divergence -2.391 , and the Kaplan-Yorke dimension is 2.08 . The positive largest Lyapunov exponent confirms the presence of chaotic dynamics. The three-dimensional attractor further highlights the bounded yet aperiodic evolution of the system states, which is a typical signature of chaos.

IV. REFERENCE TRAJECTORY GENERATION AND TRACKING ERROR FORMULATION

A realistic wind excitation is essential for evaluating the dynamic performance of a DFIG-based wind energy conversion system. In this work, the wind speed is modeled as the combination of a mean component, several low-frequency harmonic components, and a stochastic turbulence term, yielding a physically representative turbulent wind signal. The wind-speed model is expressed as

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (11)$$

here V_{mean} denotes the average wind speed, A_k and f_k are the amplitudes and frequencies of the harmonic components, and $v_{turb}(t)$ is a smoothed Gaussian noise process representing atmospheric turbulence.

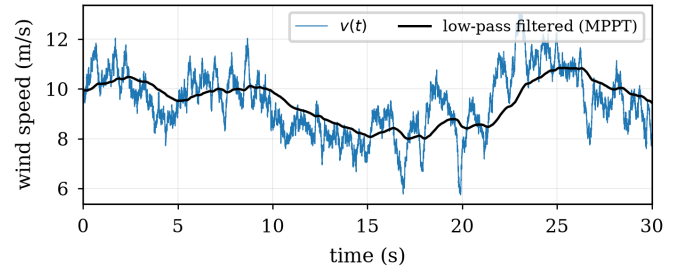


Fig. 2. Wind-speed trajectory.

Fig. 2 shows the generated wind-speed profile used to excite the DFIG system. The signal combines deterministic low-frequency variations with stochastic turbulence, providing a realistic representation of wind conditions. This complex excitation plays a key role in revealing the nonlinear and potentially chaotic behavior of the system.

Based on this profile, the aerodynamic power reference is computed from the standard wind-power expression

$$P_{ref}(t) = \frac{1}{2} \rho \pi R^2 C_p v^3(t) \quad (12)$$

where ρ is the air density, R the turbine radius, and C_p the optimal power coefficient. With the stator flux aligned with the q axis (Section III), the electromagnetic torque and the stator

active power are governed by the d -axis rotor current, leading to the reference

$$i_{dr,ref}(t) = -\frac{2}{3} \frac{P_{ref}(t)}{V_s} \frac{L_s}{L_m} \quad (13)$$

The corresponding measured active power is

$$P_m(t) = -\frac{3}{2} \frac{L_m}{L_s} V_s i_{dr}(t) \quad (14)$$

The reference for the q -axis rotor current is derived from the stator reactive-power expression under stator-flux orientation,

$$i_{qr,ref}(t) = -\frac{2L_s}{3L_m V_s} \left(Q_s(t) - \frac{3V_s^2}{2\omega_s L_s} \right) \quad (15)$$

and, by imposing $Q_s(t) = 0$, simplifies to the constant operating point

$$i_{qr,ref} = \frac{V_s}{\omega_s L_m} = \frac{\psi_s}{L_m} \quad (16)$$

which ensures unity power-factor operation.

The reference rotor speed is obtained from the optimal tip-speed ratio, giving

$$\omega_{r,ref}(t) = \frac{G \lambda_{opt}}{R} v(t) \quad (17)$$

The complete reference trajectory is therefore

$$x_r(t) = \begin{bmatrix} i_{dr,ref}(t) \\ i_{qr,ref}(t) \\ \omega_{r,ref}(t) \end{bmatrix},$$

and the corresponding tracking errors are defined as

$$e(t) = x(t) - x_r(t), \quad e_i(t) = x_i(t) - x_{r,i}(t), \quad i = 1, 2, 3.$$

A common practice in addressing the control problem of wind turbines is to use a linearization approach. However, due to the stochastic operating conditions and the inevitable uncertainties inherent in the system, such control methods come at the price of poor system performance and low reliability. Hence the need for nonlinear and robust control to take into account these control problems.

V. HYBRID CONTROL OF A NOVEL CHAOTIC DFIG-BASED SYSTEM

The controlled system takes the form

$$\begin{aligned} \dot{x}_1 &= a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + a_3 u_{ds} + a_4 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + a_3 u_{qs} + a_4 u_2, \\ \dot{x}_3 &= a_6 x_1 - a_7 x_3 - a_8, \end{aligned} \quad (18)$$

where $x = [x_1, x_2, x_3]^T = [i_{dr}, i_{qr}, \omega_r]^T$ and the two control inputs $u_1 = u_{dr}$ and $u_2 = u_{qr}$ are the rotor voltages imposed by the rotor-side converter. The speed equation has no direct input: ω_r is controlled through the electromagnetic torque, i.e., through i_{dr} . With the parameters of Table I, (18) reads

$$\begin{aligned} \dot{x}_1 &= -0.0433 x_1 + (\omega_s - x_3) x_2 - 320.83 x_3 \\ &\quad + 1.0079 \times 10^5 + 183.76 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 - 0.0433 x_2 + 0.1946 + 183.76 u_2, \\ \dot{x}_3 &= 0.3360 x_1 - 2.3044 x_3 + 727.20. \end{aligned}$$

In open loop ($u_1 = 2.539$ V, $u_2 = -2.936$ V), (18) reduces to the chaotic system (10). The objective is to force the nonlinear chaotic dynamics to follow a desired trajectory $x_d(t)$, constructed from wind-profile-based references for i_{dr} , i_{qr} , and ω_r . The tracking error is defined as $e(t) = x(t) - x_d(t)$, and its stabilization ensures both chaos suppression and accurate MPPT behavior under turbulent wind fluctuations.

The overall control structure implementing this hybrid strategy combining Model Predictive Control with nonlinear feedback compensation is illustrated in Fig. 3, where the proposed controller interacts with the turbine dynamics and the MPPT block to achieve robust trajectory tracking under variable wind conditions.

This hybrid framework harnesses the strengths of both techniques to overcome their individual limitations and improve overall system performance. Predictive control [25]–[30] is particularly valued for its ease of configuration and implementation, which makes it a widely adopted technique for controlling chaotic systems. Its efficiency, robustness, and straightforward design process enable effective management of complex dynamics with relatively low computational effort. Feedback control excels in stabilizing inherently unstable and unpredictable chaotic dynamics by continuously adjusting the control input in real time based on the system's current state. Additionally, feedback control [31]–[33] offers significant adaptability, allowing synchronization or steering of chaotic trajectories toward desired states, which is crucial for practical applications such as cryptography, robotics, and biological systems. By integrating these two approaches, the proposed hybrid control strategy capitalizes on the simplicity and robustness of predictive control alongside the real-time adaptability and energy efficiency of feedback control, thereby providing a more comprehensive solution for managing chaotic systems.

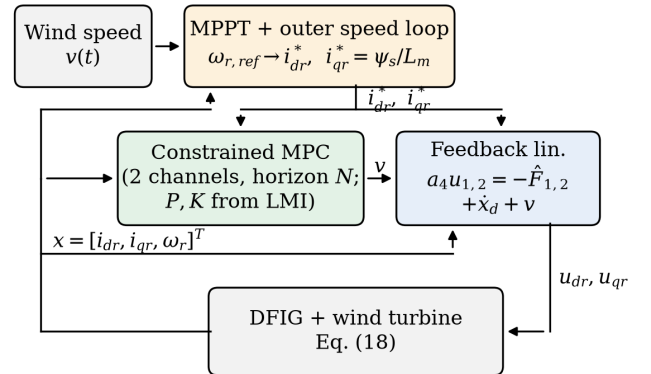


Fig. 3. Proposed hybrid MPC–nonlinear feedback control architecture: outer speed loop and inner current loops acting on the rotor voltages $u_1 = u_{dr}$, $u_2 = u_{qr}$.

A. Outer Speed Loop

Since $T_e \propto i_{dr}$, the speed is controlled through the reference of i_{dr} . With $e_\omega = \omega_r - \omega_{r,ref}$ and the nominal model,

$$\dot{i}_{dr}^* = \frac{\dot{\omega}_{r,ref} + a_7 \omega_{r,ref} + a_8 - \lambda e_\omega - \lambda_I \int e_\omega dt}{a_6}, \quad (19)$$

where $a_8 = n_p T_L / J$ uses the load (or aerodynamic) torque. Substituting (19) into the third line of (18) gives

$$\dot{e}_\omega = -(\lambda + a_7) e_\omega - \lambda_I \int e_\omega dt + a_6 (i_{dr} - i_{dr}^*),$$

an exponentially stable linear system driven by the inner tracking error; hence $e_\omega \rightarrow 0$ once $i_{dr} \rightarrow i_{dr}^*$ (cascade argument). The inner reference is $x_d = [i_{dr}^*, i_{qr,ref}]^T$ with $i_{qr,ref} = \psi_s / L_m$ from (16).

B. Inner Loop: Nonlinear Feedback Compensation

Write the first two lines of (18) as $\dot{x}_{1,2} = F_{1,2}(x) + a_4 u_{1,2}$ and let $\hat{F}_{1,2}$ be the nominal model. The control law

$$u_{1,2} = \frac{1}{a_4} \left(-\hat{F}_{1,2}(x) + \dot{x}_d + v \right) \quad (20)$$

gives $\dot{e} = v + \Delta$, with $e = x_{1,2} - x_d$ and $\Delta = F_{1,2} - \hat{F}_{1,2}$ ($\Delta = 0$ nominally). The bilinear couplings $(\omega_s - x_3)x_{1,2}$ responsible for chaos are cancelled, leaving two decoupled integrators driven by v .

C. Model Predictive Control

For each channel $i \in \{1, 2\}$, the error is augmented with its integral, $\xi_i = [e_i, \int e_i dt]^T$, and sampled with period T_s :

$$\xi_i(k+1) = A \xi_i(k) + B v_i(k), \quad A = \begin{bmatrix} 1 & 0 \\ T_s & 1 \end{bmatrix}, \quad B = \begin{bmatrix} T_s \\ T_s^2/2 \end{bmatrix}. \quad (21)$$

At each instant k , the MPC solves the quadratic program

$$\begin{aligned} \min_{v_i(\cdot|k)} \quad & \sum_{j=1}^N \|\xi_i(k+j|k)\|_Q^2 + \|\xi_i(k+N|k)\|_P^2 \\ & + r \sum_{j=0}^{N-1} v_i(k+j|k)^2 \\ \text{s.t. (21),} \quad & |u_i(k+j|k)| \leq \bar{u}, \end{aligned} \quad (22)$$

with prediction horizon N , weights $Q = Q^T > 0$, $r > 0$, and the converter voltage limit $\bar{u}$ (via (20), a bound on v_i). Only the first optimal move is applied and the problem is solved again at $k+1$ (receding horizon); the integral is frozen when the limit is reached.

Theorem 1: Let $X = X^T > 0$ and Y satisfy the LMI

$$\begin{bmatrix} X & (AX + BY)^T & XQ^{1/2} & \sqrt{r}Y^T \\ AX + BY & X & 0 & 0 \\ Q^{1/2}X & 0 & I & 0 \\ \sqrt{r}Y & 0 & 0 & 1 \end{bmatrix} \succeq 0, \quad (23)$$

and set $P = X^{-1}$, $K = YX^{-1}$. Then $(A + BK)^T P (A + BK) - P + Q + rK^T K \preceq 0$, and if (22), augmented with the terminal constraint $\xi_i(k+N|k) \in \Omega = \{\xi^T P \xi \leq \alpha\}$ (α

such that $K\xi$ respects the limits in Ω), is feasible at $k=0$ with $\Delta = 0$, it remains feasible and $\xi_i = 0$ is asymptotically stable.

Proof: The Schur complement of (23) gives $X - (AX + BY)^T X^{-1} (AX + BY) - XQX - rY^T Y \succeq 0$; pre- and post-multiplying by P with $Y = KX$ yields the stated inequality. Thus $\xi^T P \xi$ is a Lyapunov function for $v = K\xi$ and Ω is invariant. By the standard terminal-cost argument, the optimal cost decreases by at least $\|\xi_i(k)\|_Q^2 + r v_i(k)^2$ at each step, which proves recursive feasibility and asymptotic stability. ■

Remarks: The maximal-volume solution of (23) is the Riccati solution, so without active constraints the MPC returns exactly $v_i = K\xi_i$: a single gain $K = [k_e, k_I]$ from the LMI replaces the previous matrices K_1 , K_2 and the diagonal K . Setting $k_I = 0$ and ignoring constraints gives the pure feedback-linearization law $v = -k_e e$ ("FL only"); removing all model terms gives the classical cascaded PI vector control (speed PI $i_{dr}^* = -(\lambda e_\omega + \lambda_I \int e_\omega)/a_6$ and current PIs $v = K\xi$). Both baselines use the same gains, so the comparison is fair.

VI. NUMERICAL SIMULATIONS

All simulations integrate (18) with a fourth-order Runge-Kutta scheme (step $T_s/4$). The controller uses $T_s = 1$ ms, $N = 10$, $Q = \text{diag}(1, 3)$, $r = 10^{-6}$ (currents scaled by 100 A), which gives $K = [-619.2, -1069.5]$ and an LMI residual below 10^{-15} ; the outer loop uses $\lambda = 10$ s $^{-1}$, $\lambda_I = 25$ s $^{-2}$, and the rotor voltages are limited to $|u_{dr}|, |u_{qr}| \leq 200$ V.

A. Chaos Suppression

The system starts from $x(0) = (0.3, 0.6, 3.6)^T$ with the open-loop voltages of Table I and evolves on the chaotic attractor. At $t = 10$ s the controller is activated with the target $\omega_r^* = 1.05 \omega_s = 329.9$ rad/s and $i_{qr}^* = \psi_s / L_m = 22.1$ A ($Q_s = 0$); the corresponding equilibrium $i_{dr}^* = 98.0$ A is held with $u_{dr} = 29.3$ V and $u_{qr} = -8.4$ V.

TABLE II
CHAOS SUPPRESSION: IAE OF THE SPEED ERROR AND SETTLING TIME t_s TO A 1% BAND

Metric	FL-MPC	FL only	PI
IAE (rad)	3.63	3.63	2.91
t_s (s)	0.13	0.13	0.16

Fig. 4 shows that the chaotic oscillations disappear as soon as the controller is activated: the speed reaches the positive super-synchronous target and the rotor currents settle at their equilibrium values, using only the rotor voltages. Table II shows that all three controllers suppress chaos without steady-state error. During the first instants the voltages reach the 200 V limit for all controllers. In this scenario the cascaded PI gives a lower IAE, while FL-MPC settles faster (0.13 s versus 0.16 s); FL-MPC and FL only coincide, since the model error seen by the current loops is negligible.

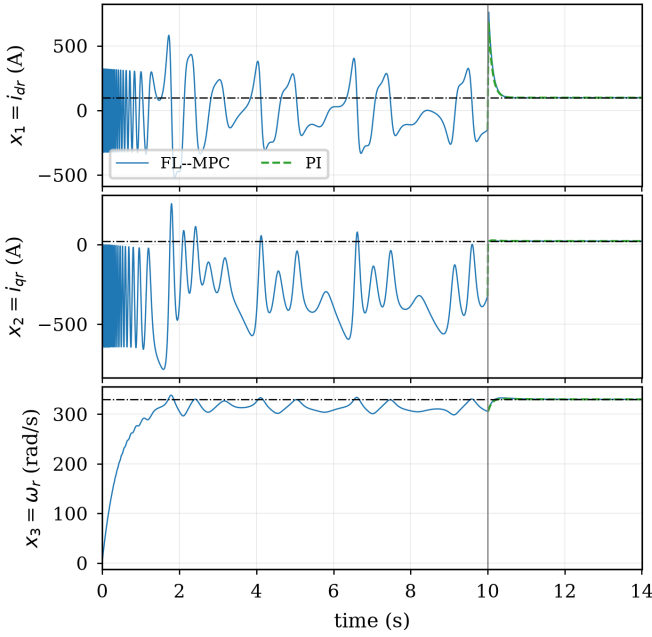


Fig. 4. Time evolution of the state variables before and after control activation at $t = 10$ s (dash-dot: target).

B. MPPT Tracking

For normal operation, the turbine torque $T_L = -T_a(v, \omega_r)$ of (3) replaces the constant load, with $R = 20$ m, $\lambda_{opt} = 8.1$, $C_{p,max} = 0.48$ and $G = 86$ in (17) (pole pairs included), and the damping is set to its nominal value $D = 0.1$ N·m·s/rad. The machine is driven by the wind of Fig. 2 (mean 9 m/s); $\omega_{r,ref}$ follows the low-pass filtered wind (time constant 2 s), so that $\omega_{r,ref}$ stays between 279 and 378 rad/s and $P_{s,ref}$ between 142 and 478 kW.

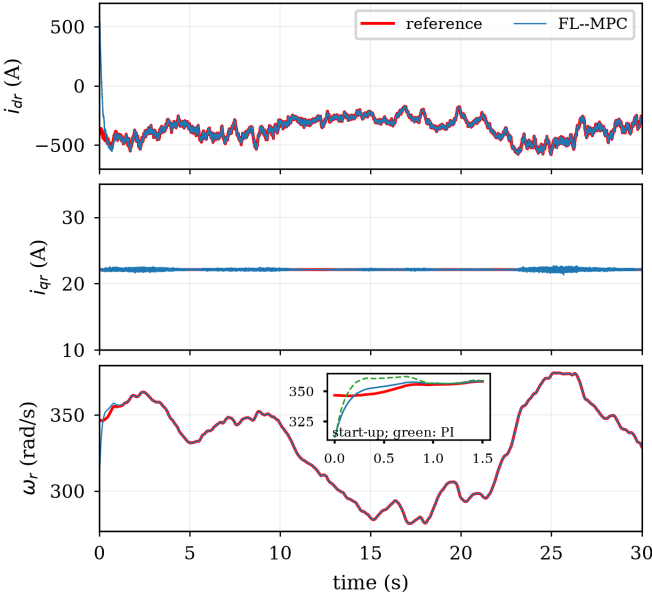


Fig. 5. Tracking performance of the state variables i_{dr} , i_{qr} , and ω_r under the proposed hybrid control scheme (inset: start-up, green: PI).

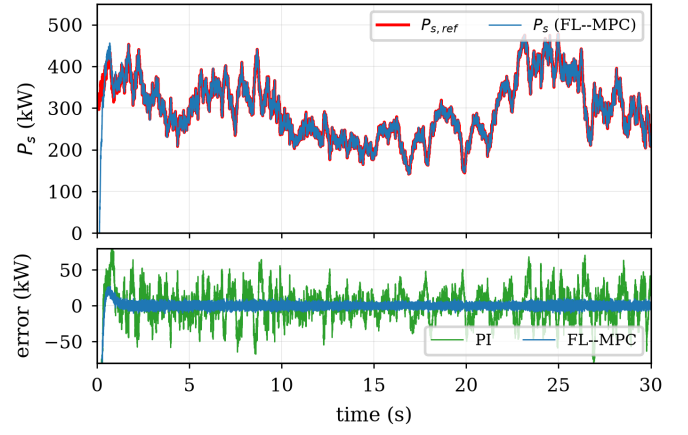


Fig. 6. Active power tracking performance: stator power P_s and MPPT reference $P_{s,ref}$ (FL-MPC), and power errors of FL-MPC and PI.

TABLE III
MPPT TRACKING ($t \geq 5$ s): RMS ERRORS

RMSE	FL-MPC	FL only	PI
ω_r (rad/s)	0.008	0.008	1.161
P_s (kW)	2.70	2.70	20.63
i_{dr} (A)	3.28	3.28	25.08

Figs. 5 and 6 show that i_{dr} , i_{qr} , ω_r and the stator power follow their MPPT references despite turbulence. Starting from $0.9\omega_{r,ref}$, the speed enters a 2% band in 0.13 s with FL-MPC versus 0.75 s with PI. Table III shows that the speed error of FL-MPC is about 146 times and the power error about 7.7 times lower than with PI. FL-MPC and FL only perform identically in tracking, because the voltage limits are active only during start-up. The specific benefit of the MPC is therefore the explicit, stability-guaranteed handling of the converter limits.

VII. CONCLUSION

This paper investigated the nonlinear dynamics of a DFIG-based wind energy conversion system with a physically consistent parameter set. The analysis of the equilibria, of the dissipativity and of the Lyapunov exponents confirmed the presence of a chaotic attractor under perturbed parameters. A hybrid controller acting only on the rotor voltages was proposed: an outer speed loop generates the torque-current reference, nonlinear feedback compensation cancels the coupled current dynamics, and a constrained MPC, whose terminal weight and gain are obtained from an LMI, handles the converter voltage limits with guaranteed nominal stability. Simulations show chaos suppression without steady-state error, and MPPT tracking under turbulent wind with speed and power errors several times lower than a classical cascaded PI controller. Future work will address converter and grid dynamics, low-voltage ride-through, and experimental validation.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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